

SMART AMBULANCE TRAFFIC SIGNAL SYSTEM

EVIDENCE & EXISTING TECHNOLOGIES

This document presents research evidence supporting the need for a Smart Ambulance Traffic Signal System. It examines measured ambulance delays, emergency-service reachability, changing traffic conditions, existing emergency-vehicle priority systems, connected-vehicle communication and dynamic ambulance routing. The evidence is used to identify the limitations of signal-only solutions and establish the technical basis for an integrated emergency-corridor system.

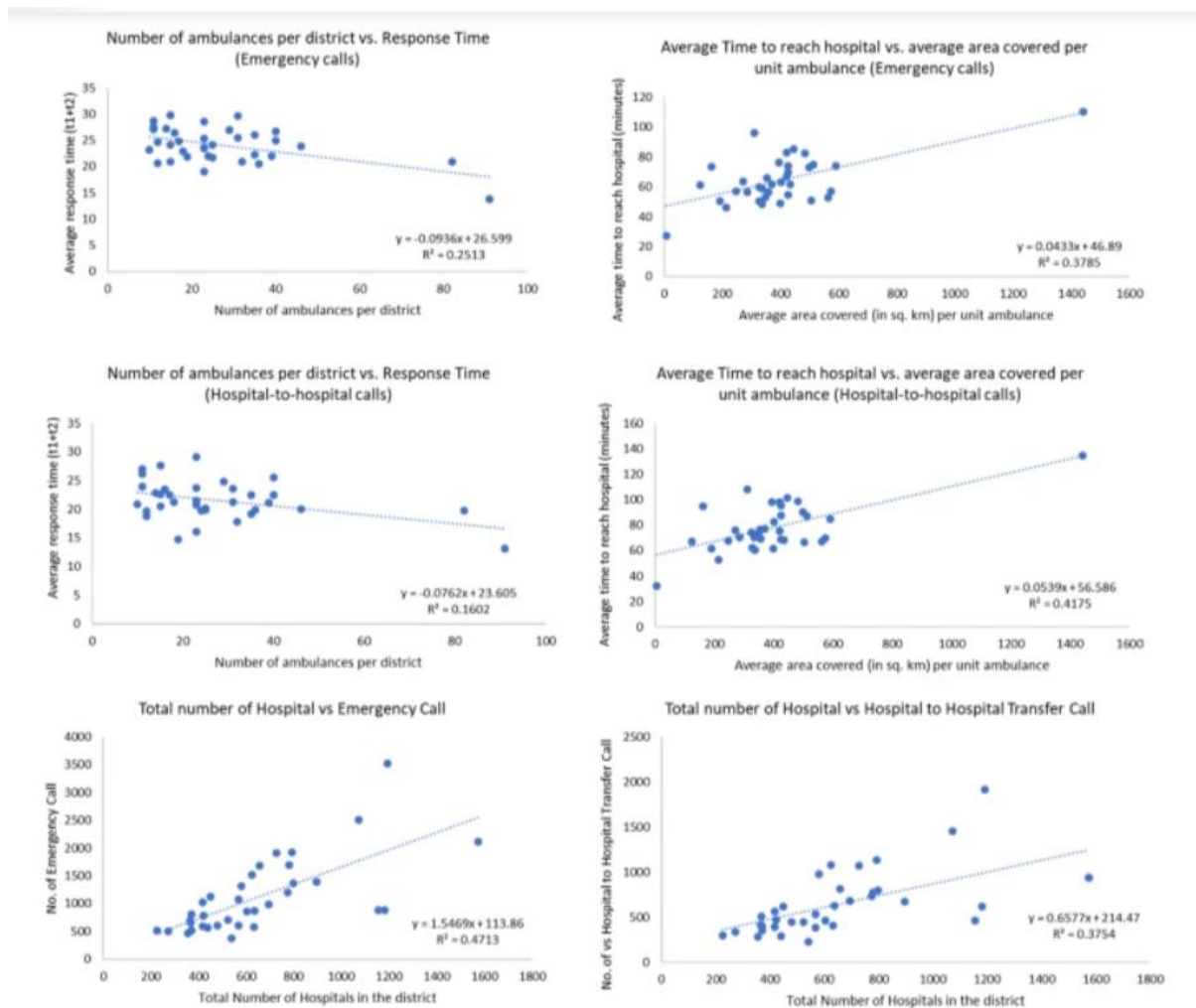
EVIDENCE 1 — AMBULANCE JOURNEYS CONTAIN MULTIPLE SOURCES OF DELAY

Study:

A peer-reviewed *Scientific Reports* study analysed **38,823 emergency medical service records across all 36 districts of Maharashtra, India**, for November 2022.

Stage	Average time
Preparation	3.53 min
Base → Scene	23.15 min
Scene Assessment	12.12 min
Scene → Hospital	39.41 min
Hospital Handover	10.85 min
Return	41.89 min

Ambulance response is a **multi-stage emergency process**. Delay does not occur only at traffic signals. Both the journey to the patient and the journey from the patient to the hospital contribute to emergency response time.



This graph shows the response time of ambulances in Maharashtra

Source:

Scientific Reports — Nature

Study: Maharashtra Emergency Medical Services

Year: 2023

Dataset: 38,823 EMS records

Coverage: 36 districts of Maharashtra

<https://www.nature.com/articles/s41598-023-48713-1>

EVIDENCE 2 — AMBULANCE REACHABILITY AND TRAUMA TRANSPORT REMAIN CHALLENGING

Study:

A **survey-based study** investigated ambulance reachability and transportation of injured people to trauma facilities following road accidents on **Indian National Highways**. The study included **225 participants**, including accident victims, relatives, witnesses and people involved in rescue activities.

Finding	Reported result
Participants surveyed	225
Successfully contacted ambulance service	~90%
Ambulance reached accident scene	~72–78%
Ambulance arrival on core highways	~25–35 min
Transport to nearby hospital	~15–25 min

The survey indicates that **successfully contacting an ambulance does not guarantee that the ambulance will reach the accident scene quickly**. The subsequent journey from the accident scene to a medical facility can also add significant time.



Image shows data about responses from ambulances

Source:

Study: Road accidents on Indian National highways, ambulance reachability and transportation of injured to trauma facility: Survey-based introspection of golden hour

Year: 2024

Study type: Survey-based study

Participants: 225

Study area: Indian National Highways

<https://pubmed.ncbi.nlm.nih.gov/38605810/>

EVIDENCE 3 — TRAFFIC CONDITIONS MAKE AMBULANCE RESPONSE TIMES VARIABLE

Study:

A 2023 study, “**Capturing delays in response of emergency services in Delhi,**” investigated delays in emergency-service response in the congested transportation network of **Delhi, India**. The study examined variation in travel times between different locations and considered dispatch delays when planning ambulance locations.

The researchers used a **Gaussian Mixture Model (GMM)** to represent travel-time variability and developed an ambulance-location planning approach based on this variation.

Finding	Study result
Study area	Delhi
Travel time	Varies with traffic conditions
Modelling method	Gaussian Mixture Model (GMM)
Demand-site coverage achieved in model	85% within 9 min*

*This 85% figure is the result of the study's specific modelling scenario and **is not a guaranteed performance value for our system**.

The study shows that emergency travel time is **not constant throughout the day**. Therefore, planning emergency response using only an average or fixed travel time may not be sufficient in a congested urban environment.

Source:

Study: Capturing delays in response of emergency services in Delhi

Authors: Wajid & Nizamuddin

Year: 2023

Journal: *Socio-Economic Planning Sciences*

Study area: Delhi, India

Method: Gaussian Mixture Model for travel-time variability

<https://www.sciencedirect.com/science/article/abs/pii/S0038012123000435>

EVIDENCE 4 — INDIA'S NATIONAL AMBULANCE FRAMEWORK SUPPORTS GPS, GIS AND INTELLIGENT DISPATCH

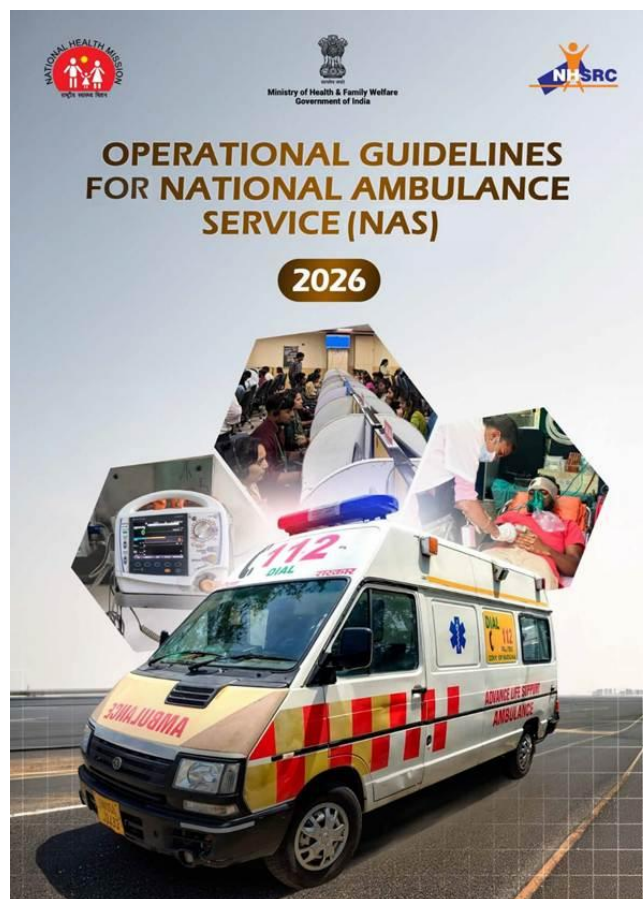
Study / Government Framework:

The **Ministry of Health & Family Welfare, Government of India**, published the **Operational Guidelines on National Ambulance Services (NAS), 2026**. The guidelines describe a national framework for improving emergency medical transport and ambulance coordination.

The framework supports the use of **GPS-enabled ambulance tracking, digital call management, intelligent dispatch, real-time monitoring and GIS-based information** for emergency medical services.

Technology / Capability	Purpose in ambulance services
GPS tracking	Monitor ambulance location
Digital call management	Manage emergency calls
Intelligent dispatch	Select and assign available ambulances
Real-time monitoring	Monitor ambulance operations
GIS mapping	Map hospitals, ambulance stations and emergency locations
Healthcare information	Identify suitable healthcare facilities
Traffic & geographic data	Support ambulance deployment and planning

The guidelines also describe mapping information such as **healthcare facilities, referral centres, ambulance stations, accident-prone/high-risk locations, bed availability and critical-care preparedness.**



This diagram shows how GPS, GIS, emergency-call information and intelligent dispatch can support ambulance coordination.

Source:

Government: Ministry of Health & Family Welfare, Government of India

Document: Operational Guidelines on National Ambulance Services (NAS), 2026

Year: 2026

Key areas: GPS tracking, digital call management, intelligent dispatch, GIS and real-time monitoring

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2279073&lang=2®=48>

EVIDENCE 5 — SINGAPORE'S TRAFFIC PRIORITY SYSTEM TARGETS THE FINAL MILE

Study / Existing System:

Singapore Civil Defence Force (SCDF), together with the Land Transport Authority (LTA) and Home Team Science & Technology Agency (HTX), developed a **Traffic Priority System (TPS)** for emergency ambulances.

The system was first implemented near Ng Teng Fong General Hospital in July 2024 and was progressively expanded in 2025 to junctions near **seven additional public hospitals**. It provides traffic-signal priority during the **final mile of casualty conveyance**, typically within about **200 metres of TPS-enabled junctions near hospitals**.

Finding	Reported result
First implementation	July 2024
Public hospitals covered by end of 2025	8
Activations by 31 December 2025	>2,500
Average time savings	Up to 1 min 57 sec per conveyance
Priority location	Final mile near hospitals
Typical junction distance	Within ~200 m

The concentration of the system near hospitals shows that **traffic-signal priority can be particularly useful at critical final intersections where even a short delay can affect the ambulance's arrival at the medical facility**. It also demonstrates that emergency traffic priority can be deployed selectively at locations where it provides a measurable operational benefit, rather than requiring every road junction to have the same system.

Source:

Organization: Singapore Civil Defence Force (SCDF)

Partners: Land Transport Authority (LTA), Home Team Science & Technology Agency (HTX)

System: Traffic Priority System (TPS)

Year: 2025

Coverage: Junctions near 8 public hospitals by end of 2025

Reported activations: More than 2,500

Reported time savings: Up to 1 min 57 sec per conveyance

<https://www.scdf.gov.sg/docs/default-source/media-room-%28publications%29/annual-statistics/scdf-annual-statistics-on-ems-fire-enforcement-2025.pdf>

EVIDENCE 6 — UNITED STATES SYSTEMS SHOW HOW EMERGENCY VEHICLE PRE-EMPTION CAN AFFECT THE WHOLE TRAFFIC NETWORK

Study / Existing Technology:

The **Federal Highway Administration (FHWA)** documents emergency-vehicle signal pre-emption systems used in the United States. These systems detect approaching emergency vehicles and temporarily alter normal traffic-signal operation to provide right-of-way.

Emergency vehicles can be detected using several technologies, including **optical signals, sirens, pavement loops, radio transmission and GPS-based communication**. When an emergency vehicle is detected, the signal controller can change or extend the signal phase to provide priority to the vehicle's direction of travel.

Technology / Function	Purpose
Emergency-vehicle detection	Detect approaching ambulance/emergency vehicle
Radio / optical / GPS communication	Send priority request
Signal controller	Process the priority request
Signal pre-emption	Provide right-of-way
Coordinated signals	Support progression through multiple intersections
Post-pre-emption recovery	Return traffic signals to normal coordination

FHWA also notes that emergency pre-emption can temporarily **disrupt normal traffic flow**. Congestion can occur or become worse before traffic returns to normal operation. In one documented Houston deployment involving 22 intersections, average emergency-vehicle travel time decreased by **16–23%**, although effectiveness varied between deployments.

Connected-vehicle research has also examined using **V2V and V2I communication** so an emergency vehicle can broadcast information such as its location, route and destination to vehicles and infrastructure along its path. This can support safer coordination rather than relying only on individual drivers noticing the ambulance.

Source:

Organization: Federal Highway Administration (FHWA), U.S. Department of Transportation

Technology: Emergency Vehicle Signal Pre-emption / V2I

Key areas: Emergency-vehicle detection, signal pre-emption, coordinated traffic control, V2V/V2I

Deployment evidence: Houston, Texas — 22 intersections

Reported travel-time reduction: 16–23% in that deployment

<https://www.fhwa.dot.gov/publications/research/operations/its/06108/03.cfm>

https://ops.fhwa.dot.gov/trafficanalysistools/tat_vol4/app_n.htm

EVIDENCE 7 — INDIA HAS ALREADY TRIALED V2X-BASED GREEN CORRIDORS FOR EMERGENCY VEHICLES

Study / Technology Trial:

The **Telecommunications Engineering Centre (TEC), Government of India**, documents an Indian trial of a **V2X-based green corridor for emergency vehicles**.

The trial integrated Variable Message Signs (VMS) with existing and new traffic-signalling systems using V2X infrastructure. According to the TEC report, the system could **change a red signal to green and extend a green signal cycle** to provide priority passage for emergency vehicles.

Trial feature	Function
V2X infrastructure	Communication between vehicles and road infrastructure
Emergency-vehicle detection	Identify the approaching emergency vehicle
Signal-cycle change	Change red to green for priority movement
Green extension	Extend green time for emergency passage
Variable Message Signs	Support traffic information/communication
Green corridor	Provide priority movement through successive traffic-control points

The report describes trials conducted in **Ahmedabad**, including a Phase-I trial in 2018 and a Phase-II trial for emergency-vehicle detection.

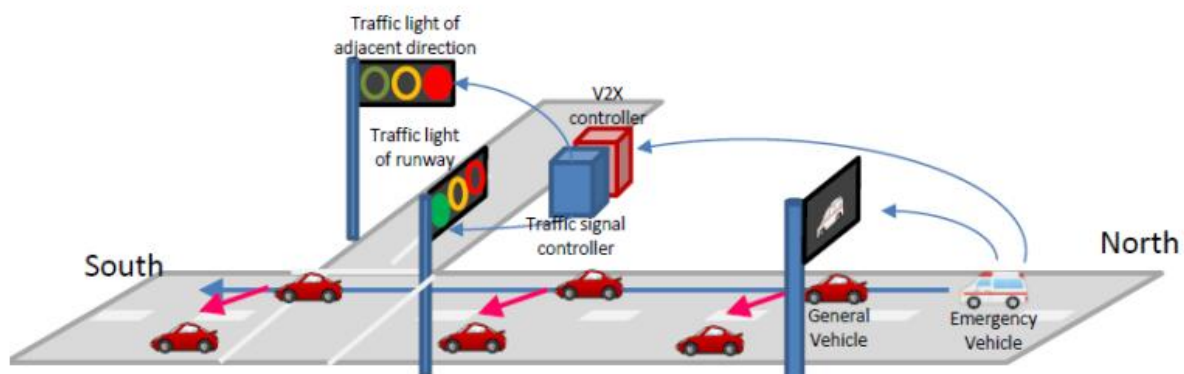


Figure 18: Trial for V2X infrastructure to enable a 'green corridor' for emergency vehicles

This is important because it shows that **the basic concept of using connected infrastructure to create a green corridor for emergency vehicles has already been demonstrated in India**.

Source:

Organization: Telecommunications Engineering Centre (TEC), Government of India

Document: *Technologies and Standards for Intelligent Transport System — Technical Report*

Section: ITS Application Examples and Experiments in India

Trial: Green Corridor for Emergency Vehicles using V2X Infrastructure

Location: Ahmedabad

Key technologies: V2X, traffic-signal control, Variable Message Signs, emergency-vehicle detection

https://www.tec.gov.in/public/pdf/M2M/TR_Technologies%20and%20Standards%20for%20Intelligent%20Transport%20System.pdf

Conclusion

The evidence shows that ambulance delays can occur throughout the emergency journey, not only at traffic signals. Existing systems already demonstrate GPS-based dispatch, signal priority, V2X communication and green-corridor technology. However, these technologies are often focused on specific parts of the journey. Therefore, our project proposes an **integrated Dynamic Emergency Corridor** that continuously coordinates routing, traffic prediction, driver alerts and traffic signals from the emergency call to the accident scene and finally to the hospital.